Risk Management Questions

Question 1

A project manager identifies a data breach risk with:
- Probability (P) = 0.4
- Cost (C) = $50,000

A mitigation strategy will reduce the probability to 0.2 at a mitigation cost of $8,000.

i) Calculate the Risk Exposure (RE) before mitigation.
ii) Calculate the new Risk Exposure (RE) after mitigation.
iii) Determine the savings from mitigation and decide if the strategy is worth applying based on the $8,000 cost.

Question 2

A software development project faces a vendor delay risk with:
- Probability (P) = 0.5
- Cost (C) = $25,000

A mitigation plan involving multiple suppliers reduces the probability to 0.3, with a mitigation cost of $6,000.

i) Calculate the Risk Exposure (RE) before mitigation.
ii) Calculate the new Risk Exposure (RE) after mitigation.
iii) Determine the savings from mitigation and decide if the strategy is worth the $6,000 cost.

Question 3

In an IoT-based healthcare project, the sensor malfunction risk is identified with:
- Probability (P) = 0.7
- Cost (C) = $40,000

A mitigation strategy that involves using backup sensors reduces the probability to 0.4, with a mitigation cost of $12,000.

i) Calculate the Risk Exposure (RE) before mitigation.
ii) Calculate the new Risk Exposure (RE) after mitigation.
iii) Determine the savings from mitigation and whether the strategy is cost-effective

given the $12,000 mitigation expense.